

Roll No.

TEE-706

B. TECH. (EE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022

UTILIZATION OF ELECTRICAL ENERGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks

1. (a) Discuss the industrial applications of electric heating. What are the advantages of electric heating ? (CO1)

OR

- (b) A resistance oven employing nichrome wire is to be operated from 230 V single-phase supply and is to be rated at 32 kW. If the temperature of the element is to be limited to 1170°C and average temperature of the charge is 500°C, find the diameter and length of the element wire : (CO1)

Radiating efficiency = 0.57, Emissivity = 0.9, Specific resistance of nichrome = 109×10^{-8} ohm-m.

P. T. O.

2. (a) Discuss conducting-bottom type Direct Arc Furnace. (CO1)

OR

- (b) An insulating material 2 cm thick and 200 cm^2 in area is to be heated by dielectric heating. The material has relative permittivity of 5 and power factor of 0.05. Power required is 400 W and frequency of 40 MHz is to be used. Determine the necessary voltage and the current that will flow through the material. If the voltage were to be limited to 700 V, what will be the frequency to get the same loss ? (CO1)

3. (a) Explain in brief how heating is done in dielectric heating. (CO1, CO2)

OR

- (b) Discuss the laws of Illumination. (CO1, CO2)

4. (a) Enumerate the various factors, which have to be considered while designing any lighting scheme. (CO2)

OR

- (b) A corridor is lighted by 4 lamps spaced 10 m apart and suspended at a height of 5 m above the centre line of the floor. If each lamp gives 200 C.P. in all directions below the horizontal, find the illumination at the point on the floor mid-way between the second and third lamps.

(CO2)

5. (a) Explain the various types of lighting schemes with relevant diagrams.

(CO2)

OR

- (b) A hall 30 m long and 12 m wide is to be illuminated and the illumination required is 50 lm/m^2 . Calculate the number, the wattage of

(3)

each unit and the location and mounting height of the units, taking a depreciation factor of 1.3 and utilization factor of 0.5, given that the outputs of the different types of lamps are as under : (CO2)

Watts	Lumens
100	1615
200	3650
300	4700
500	9950
1000	21500